What is a Solution?

A **solution** is a **homogeneous mixture** of two or more substances in which the components are evenly distributed throughout the mixture.

Solute

The substance that is dissolved. It is usually present in a smaller amount.



Example: Salt

+

Solvent

The substance that dissolves the solute. It is usually present in a larger amount.



Example: Water

→

Solution

A homogeneous mixture formed when the solute is dissolved in the solvent.



Example: Saltwater

- **Homogeneous mixture**
- Solute particles are very small (cannot be seen).
- Appear as a single phase (uniform).
- Cannot be separated by ordinary filtration.

Types of Solutions
(Based on the States of Matter)

Gas in gas

Example: Air



Solute: Other gases (e.g., carbon dioxide)

Solvent: Nitrogen

Explanation:
Different gases are uniformly mixed.

Gas in liquid

Example: Carbonated water



Solute: Carbon dioxide gas

Solvent: Water

Explanation:
Carbon dioxide gas is dissolved in water.

Liquid in liquid

Example: Vinegar



Solute: Acetic acid

Solvent: Water

Explanation:
Acetic acid is mixed with water.

Solid in liquid

Example: Saltwater



Solute: Salt (sodium chloride)

Solvent: Water

Explanation:
Salt is dissolved in water.

Solid in solid

Example: Brass



Solute: Zinc

Solvent: Copper

Explanation:
Zinc is distributed throughout copper in a solid alloy.

Solutions in Household Products

Many household products are **solutions** where the solute is dissolved in a solvent, usually water.

Vinegar



Solute: Acetic acid

Solvent: Water

Used for cooking, food preparation, and some cleaning tasks.

Rubbing Alcohol



Solute: Isopropyl alcohol (or ethanol)

Solvent: Water

Used for disinfecting surfaces and as an antiseptic (in proper concentration).

Sugar Water



Solute: Sugar (sucrose)

Solvent: Water

Used for preparing beverages and other food products.

Saltwater



Solute: Salt (sodium chloride)

Solvent: Water

Used in food preparation and some household rinsing tasks.

Food Coloring



Solute: Food coloring dye

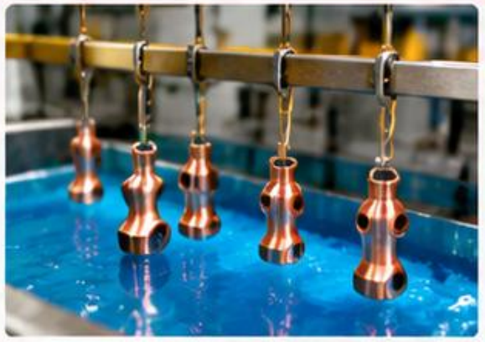
Solvent: Water

Used to add color to foods and drinks.

Solutions in Industry

Solutions are widely used in **industry** for manufacturing products, processing materials, and carrying out chemical reactions.

Electroplating



Solute: Copper(II) sulfate ($CuSO_4$)

Solvent: Water

Used to coat metal objects with a thin layer of metal for protection and aesthetic purposes.

Food and Beverage Production



Solute: Sugar, flavorings, acids, and other food additives

Solvent: Water

Used to make beverages, sauces, and other food products with the desired taste and composition.

Automotive Industry



Solute: Ethylene glycol (or propylene glycol) and additives

Solvent: Water

Used in engine coolants to regulate temperature and protect engines from freezing and overheating.

Chemical Manufacturing



Solute: Sodium hydroxide (NaOH)

Solvent: Water

Used to prepare solutions for chemical reactions and product formulation.

Pharmaceutical Industry



Solute: Active ingredients (e.g., antibiotics, vitamins)

Solvent: Water, ethanol, or other suitable solvents

Used to produce medicines and other healthcare products in solution form.

Solutions in the Environment

Natural water sources, the air, and the Earth's systems contain many solutions. Dissolved substances play important roles in ecosystems and affect the quality of our environment.

Seawater



Solute: Dissolved salts (e.g., sodium chloride, magnesium salts)

Solvent: Water

Importance: Provides essential ions for marine organisms and influences ocean circulation and climate.

Freshwater (rivers and lakes)



Solute: Dissolved minerals (e.g., calcium, magnesium, bicarbonate) and gases (e.g., oxygen, carbon dioxide)

Solvent: Water

Importance: Supports aquatic life and provides water for human use and agriculture.

Soil Water (groundwater)



Solute: Dissolved minerals (e.g., calcium, iron, nitrates)

Solvent: Water

Importance: Supplies minerals to plants and is an important source of drinking water.

Air



Solute: Different gases (e.g., oxygen, carbon dioxide, water vapor)

Solvent: Nitrogen

Importance: Provides the gases needed for life, weather, and climate processes.

Natural Waters with Pollutants



Solute: Pollutants (e.g., nitrate ions, phosphate ions, heavy metal ions)

Solvent: Water

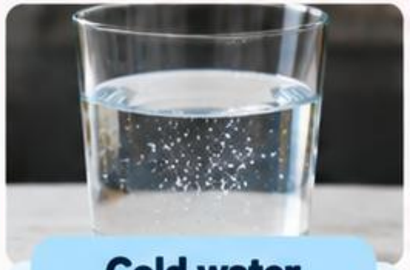
Effect: Some dissolved pollutants can harm aquatic life and reduce water quality, even if the water looks clear.

Factors Affecting the Formation of Solutions

The rate of dissolving and the amount that can dissolve (solubility) depend on several factors.

Temperature

For **many solid solutes**, increasing temperature **increases the amount that can dissolve**.



Cold water

Less sugar dissolves.



Hot water

More sugar dissolves.

Note: For gases dissolved in liquids, increasing temperature generally **decreases gas solubility**.

Stirring

Stirring helps the solute **dissolve faster** by bringing fresh solvent into contact with the solute.



Without stirring

Dissolves slowly.



With stirring

Dissolves faster.

Particle Size

Smaller particles dissolve faster because they have a greater surface area.



Larger particles (sugar cubes)

Dissolves slowly.



Smaller particles (powdered sugar)

Dissolves faster.

For Gases Dissolved in Liquids

- Increasing **temperature** generally **decreases gas solubility** (e.g., less CO_2 dissolves in warm water).
- Increasing **pressure** above the liquid generally **increases gas solubility** (e.g., CO_2 stays dissolved in a sealed bottle).

Important Reminders

- Stirring, higher temperature, and smaller particle size usually make dissolving faster (affect the **rate of dissolving**).
- These factors **do not always increase the maximum amount that can dissolve** (solubility) at a given temperature.

Salt-Water Solutions in Daily Life



Salt-water solutions are useful in everyday life, such as in **cooking** and **food preservation.**

Cooking

Salt is dissolved in water to season food and enhance its flavor.



Food Preservation

Salt-water solutions help prevent the growth of harmful microorganisms, keeping food fresh for a longer time.





Salt dissolves in water to form a **solution** that is useful in many everyday applications.

Plants and the Quality of Water

Plants need **water, sunlight, air,** and **nutrients** to grow.

Water

Helps in transport of nutrients and supports growth.

Sunlight

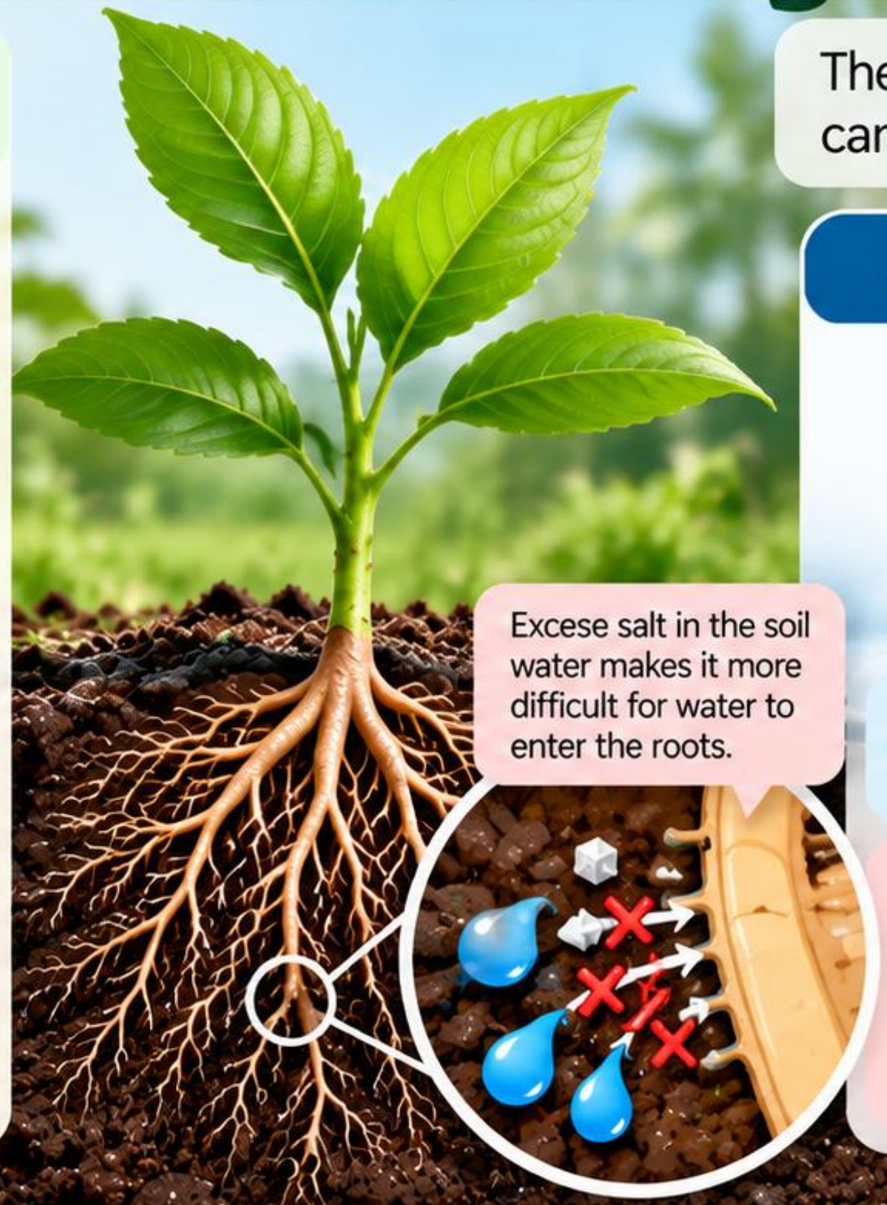
Provides energy for photosynthesis (to make food).

Air

Provides carbon dioxide for photosynthesis.

Nutrients

Needed for healthy growth and development.



The **quality of the water** they absorb can affect their **growth and survival.**

Salt dissolved in water



When salt dissolves in water, it forms a **salt solution.**

Small amounts of certain dissolved salts may provide **essential nutrients.**

However, **excessive salt** can make it **difficult for plants to absorb water** through their roots.

How Solutions of Saltwater and Wastewater Can Affect Local Ecosystems

Saltwater Solutions

When excess salt dissolves in water, it **increases the salinity** of the environment. Changes in salinity can affect the organisms that live in the water and nearby areas.

Can cause **stress** or **death** to **marine organisms** not adapted to high salinity.

High salinity can affect coastal plants such as mangroves, **reducing their growth and survival.**

Can change the types of **organisms** that can live in the area, **affecting biodiversity**.

Wastewater Solutions

Wastewater contains dissolved and suspended substances such as **chemicals**, **organic matter**, **and nutrients**. When released into the environment, these solutions can harm the water, soil, and living organisms.

Excess nutrients can cause **algal blooms**, which reduce dissolved oxygen in the water.

Lower oxygen levels can **harm or kill fish** and other aquatic organisms.

Harmful substances in wastewater can **affect the growth and survival** of aquatic plants and other living organisms.

Can also affect **human activities** such as fishing, recreation, and the use of water for daily needs.

Ways to Minimize the Negative Impact of Saltwater and Wastewater on the Environment

Saltwater Solutions

Excess salt in water can harm plants, animals, and ecosystems.

1 Protect and restore mangroves

Mangroves help buffer coastal areas from saltwater and protect habitats for marine life.



2 Prevent excessive salt entry into freshwater areas

Proper management of coastal development, irrigation, and water extraction can help maintain a healthy balance of salinity.



3 Use sustainable farming practices

Avoid overuse of salty irrigation water and use drainage systems to prevent salt buildup in soil and waterways.



Wastewater Solutions

Wastewater can contain chemicals, organic matter, and nutrients that can harm ecosystems.

1 Treat wastewater before release

Use proper treatment facilities to remove harmful chemicals, excess nutrients, and suspended particles.



2 Reduce the use of harmful chemicals

Use eco-friendly and biodegradable products at home, in industries, and in agriculture.



3 Proper waste disposal

Do not dump chemicals, oils, or untreated wastewater into rivers, lakes, or the sea.

